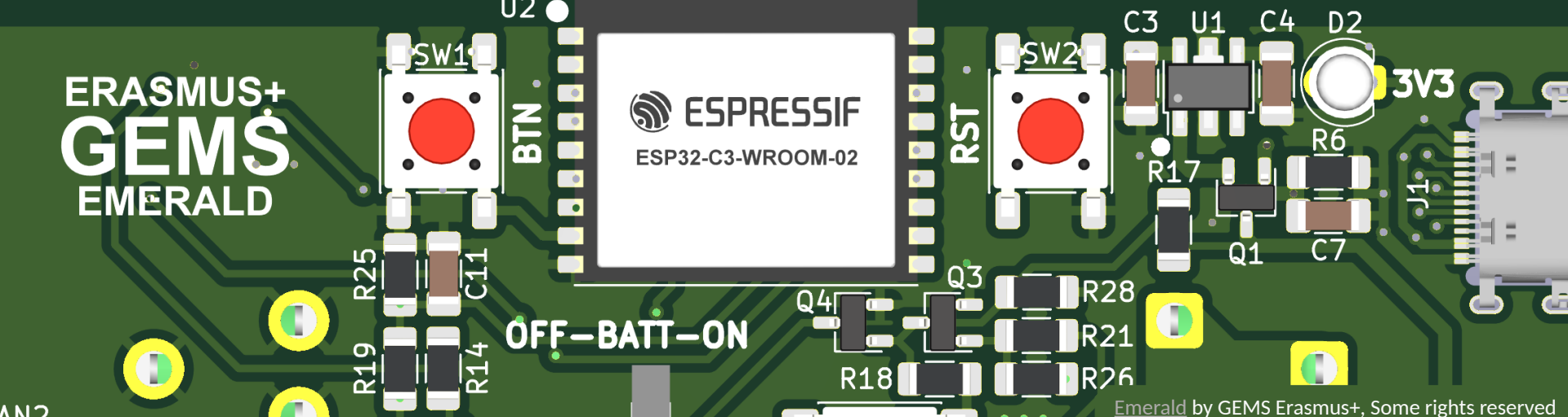


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Wirelessly controlled robot – Information flow

Building Robots: System Integration

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About

- 1 Wireless control
- 2 Technology selection
- 3 Information flow specification
- 4 Information flow implementation

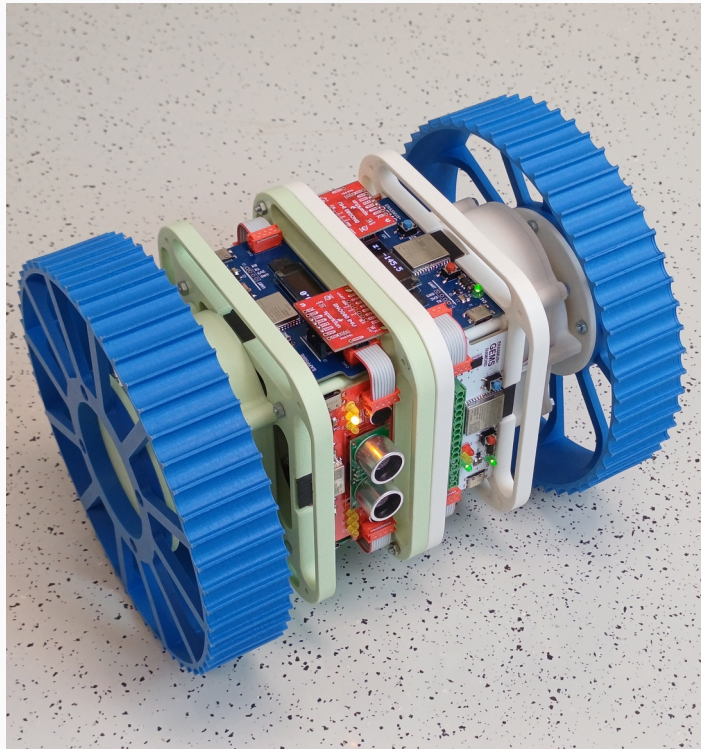
Wireless control





Foundation for wireless control

- 1 Mobile robot → autonomous movement in its environment
- 2 How to move? → make information connections
- 3 Start → manual control
Upgrade → automatic control



GEMS mobile robot setup

- 1 Module - dedicated function
→ modules must communicate
- 2 Servomotor has a battery
→ power cable not needed
- 3 Module - communication connections: CAN bus, WiFi, Bluetooth, (I2C, UART)

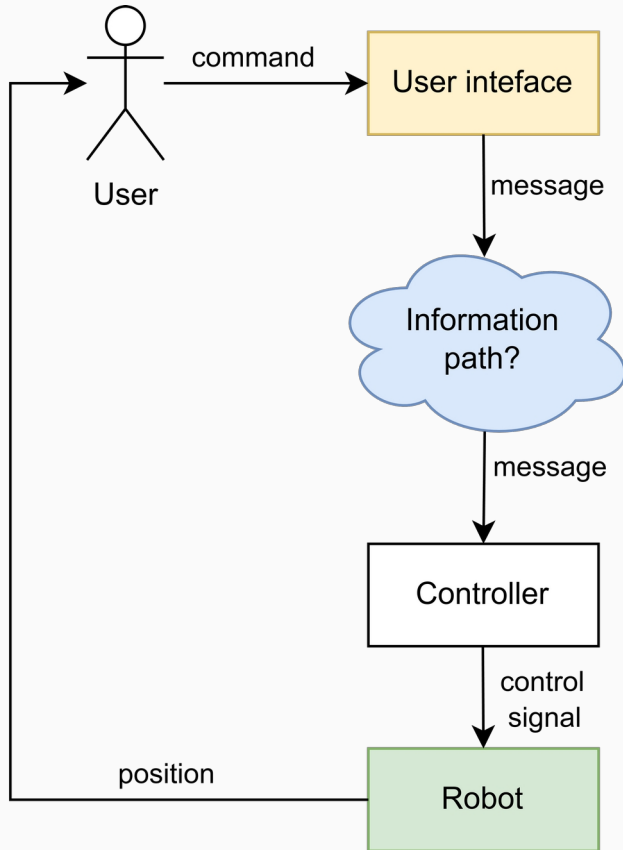


Manual wireless control

- 1 Automatic control is possible
manual control → safer operation
- 2 Wired communication is possible
→ obstacle for movement, range
- 3 Access to compatible communication devices → Smartphone and PC

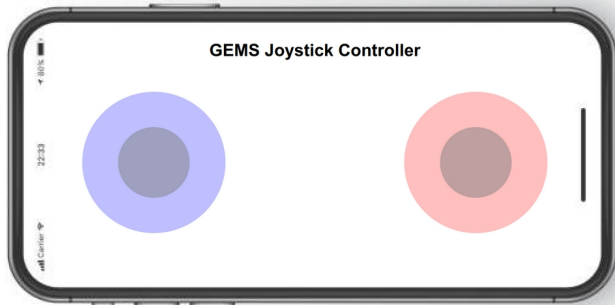
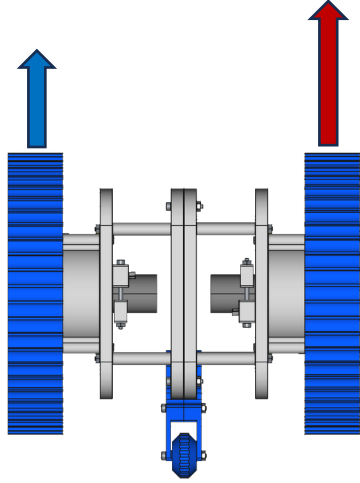
Information flow





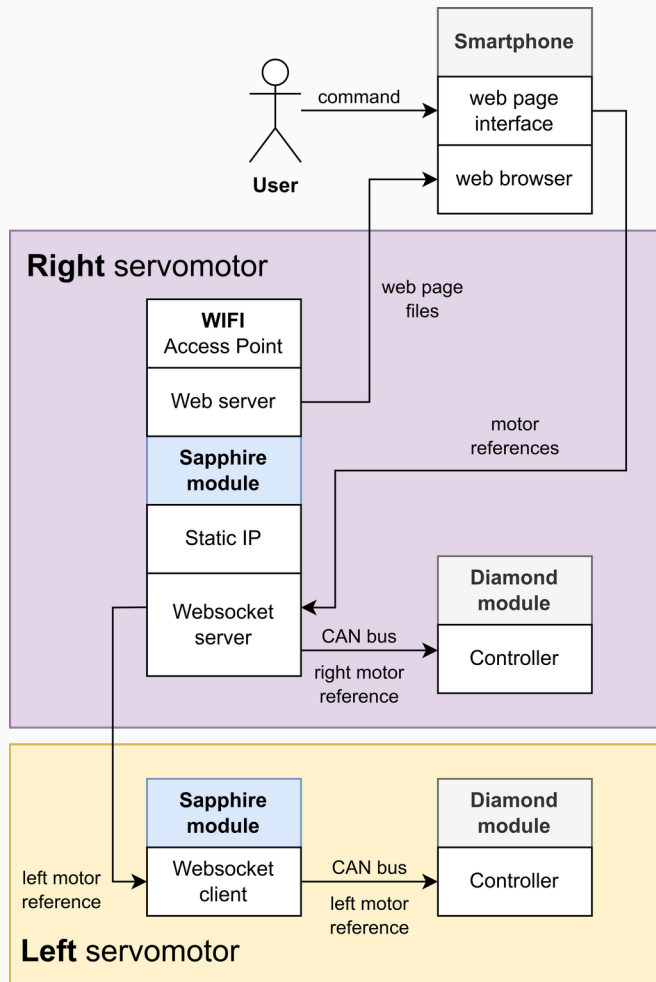
Problem

- 1 **User command:**
motor speed references
- 2 **Command target:**
DC motor control module
- 3 **Information path?**



Specification

- 1 User interface
→ web page on a smartphone
- 2 no additional infrastructure
→ simple communication paths
- 3 Responsive control
→ fast and non-blocking



Implementation

- 1 Available web technologies
html, css, js → interface libraries
- 2 WiFi Access Point (AP) and Web
server on a microcontroller
- 3 Wireless only when necessary
→ modules will use CAN bus

Task

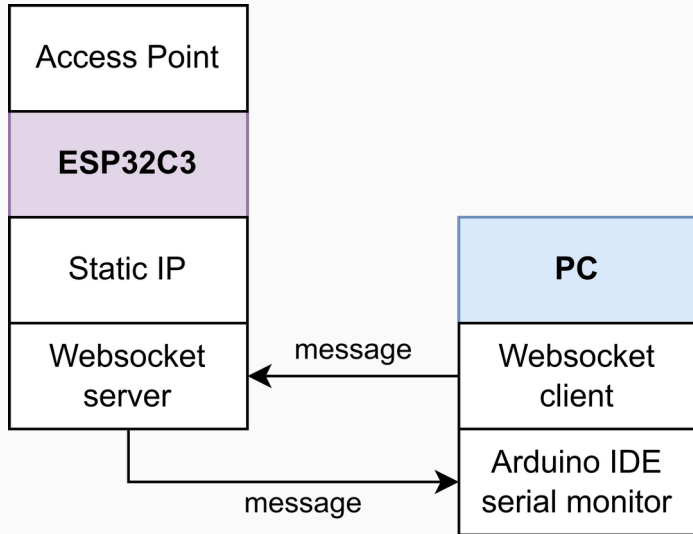


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Web socket server

Arduino program

- 1 ESP32C3/S3, Firefox browser + extension WebSocket Weasel
- 2 Access Point (AP) on microcontroller → PC connects to AP
- 3 Websocket (ws) connection ws client on PC browser → ws server on microcontroller

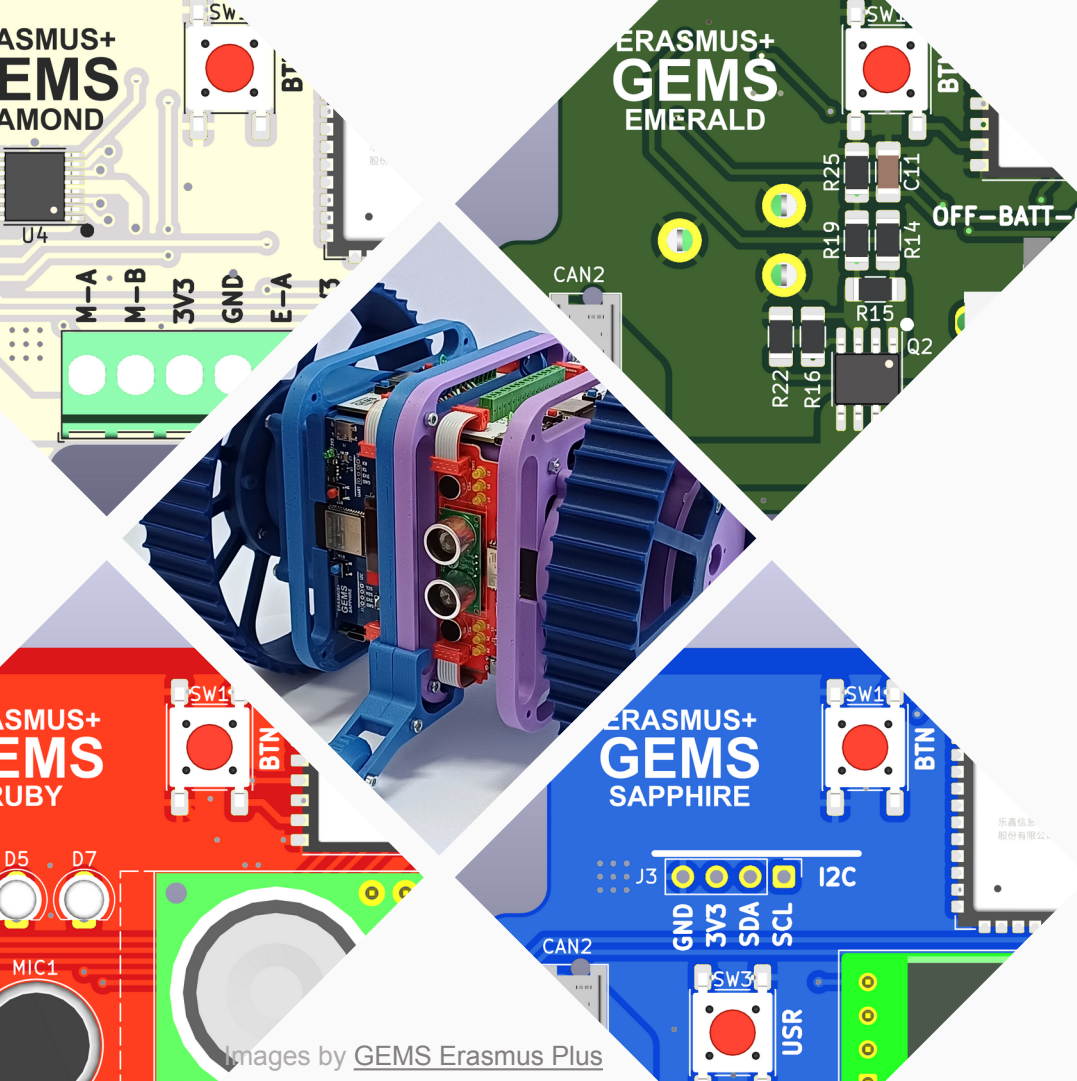
Conclusion





Wirelessly controlled mobile robot

- 1 Specify information flow early in the design process
- 2 Independant operation
no internet infrastructure
- 3 Fast/responsive, robust, safe



Thank you for watching!

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